
README FILE TO USE STATA CODE FOR "Passthrough of Emissions Costs in Electricity Markets"

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- The main file that runs the results is "master_run_code.do", which calls other Stata files (included in subfolder "stata_code").
- Code to generate results is organized in three sections:
 - 1) DATA CLEAN-UP: Reading and creating data
 - 2) SUMMARY: Summary statistics, descriptive tables and graphs
 - 3) MAIN ANALYSIS: Generates main results of the paper
 - 4) ROBUSTNESS: Regressions in additional materials documents

HOW TO RUN / SETUP:

1. Save files in a folder.
2. Open master_code_run.do in "stata_code".
3. Set \$dirpath variable to local path (main folder where all files are).

Notes: - Code is commented to only run main analysis given cleaned data sets.
- Each code can be run separately or with master_code_run file.
- Global variable \$dirpath needs to be defined.

FILES DESCRIPTION:

1) DATA CLEAN-UP

- *data_create.do***: reads raw files from the market operator and computes residual demand elasticities for the major firms in the market (raw data can be obtained at www.omie.es).
- *data_create_demand.do***: reads raw files to get measures of market outcomes (raw data can be obtained at www.omie.es or upon request).
- *data_merge.do***: reads generated data and merges it with auxiliary data sets on weather, carbon prices, etc. to create clean data sets.
- *data_create_matlab.do***: creates supply curve files to be read in matlab (numerical files). Requires large raw data available at the market operator website (www.omie.es) or upon request.
- *elasticityMarkupAnalysis.m***: computes perturbations to supply curves for first-order analysis in section III.A.

Note: Requires solver "cplex" which is offered by IBM Academics for free.

2) FIGURES

- *prelim_graphs.do*: Generates Figures 1 and 2 in the paper.

3) MAIN ANALYSIS

- *analysis_passthrough_rf.do*: generates the results for the reduced form passthrough in section II.B. (Tables 1 and 2) and appendix. Stores log files and tables in “tables” subfolder.
- *analysis_internalization_str.do*: generates the results for the structural regression in section III.A, Table 3. Stores log files and tables in “tables” subfolder.
- *analysis_markup.do**: generates markup decomposition in section III.A, Table 4. Stores log files and tables in “tables” subfolder.
Note: Requires markup dataset, which is already generated and included as part of the matlab data. A sample markup dataset can be generated using the code “elasticityMarkupAnalysis.m”.
- *analysis_rigidities.do**: looks at bidding changes at the unit and firm level.
Note: Requires large raw data available at the market operator website (www.omie.es) or upon request. A subsample of the data are included.

4) ONLINE APPENDIX

- *analysis_passthrough_robustness.do*: generates results with trends in online appendix.

ADDITIONAL NOTES AND COMMENTS:

- The main dataset used in the analysis are “data_regressions_passthrough.dta” and “data_regressions_structural.dta,” which are already cleaned up to replicate the main results in the paper and included in the subfolder “stata_data.”
- Code marked with * can be run on a subsample due to data size.
- Code marked with ** cannot be run, as it requires native raw files from the system and market operator.